

Scheduling in Kubernetes

What is Scheduling?

Definition

Scheduling is the process of deciding which Worker Node will run a Pod.

Simple Definition

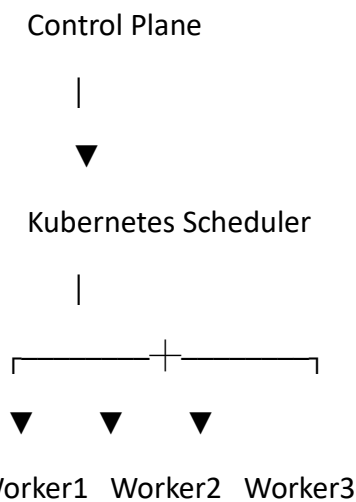
When a new Pod is created, Kubernetes must decide **where to place it**.

This decision is called **Scheduling**.

Why is Scheduling Needed?

Imagine you have **3 Worker Nodes**.

Kubernetes Cluster



Now you create a Pod.

Question:

👉 Which Worker Node should run this Pod?

The **Scheduler** makes this decision.

Real-Life Example 

Imagine a hotel.

-  Hotel = Kubernetes Cluster
-  Rooms = Worker Nodes
-  Guest = Pod
-  Receptionist = Scheduler

When a guest arrives:

The receptionist checks:

- Which room is empty?
- Which room is available?

Then assigns the guest to that room.

Similarly,

The Kubernetes Scheduler checks which Worker Node has enough resources and assigns the Pod there.

Automatic Scheduling (Default)

Normally, Kubernetes schedules Pods **automatically**.

Example:

Worker Node 1 → Busy ❌

Worker Node 2 → Free ✅

Worker Node 3 → Busy ❌

Scheduler chooses:

Worker Node 2 ✅

You don't need to do anything.

How Does the Scheduler Decide?

The Scheduler checks:

- Available CPU
- Available Memory (RAM)
- Node health
- Resource requirements of the Pod
- Scheduling rules (if any)

Then it selects the most suitable Worker Node.

Scheduling Flow

User

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Create Pod

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API Server

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Scheduler

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Chooses Best Worker Node

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Kubelet

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Pod Starts Running

Manual Scheduling

What is Manual Scheduling?

Definition

Manual Scheduling means the user decides which Worker Node will run the Pod instead of letting the Scheduler choose automatically.

Why Use Manual Scheduling?

Sometimes you want a Pod to run on a **specific Worker Node**.

Example:

- A node has special hardware (GPU).
 - A node has more storage.
 - Testing on a particular machine.
 - Running an application close to specific resources.
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How Does Manual Scheduling Work?

Instead of the Scheduler selecting the node,
you specify the node yourself using the **nodeName** field in the Pod YAML.

Example YAML

```
apiVersion: v1
```

```
kind: Pod
```

metadata:

name: nginx-pod

spec:

nodeName: worker-node-1

containers:

- name: nginx

image: nginx

Explanation

nodeName: worker-node-1

This tells Kubernetes:

"Run this Pod only on **worker-node-1**."

The Scheduler is **bypassed**.

Automatic vs Manual Scheduling

Automatic Scheduling

Create Pod

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Scheduler

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Chooses Best Node

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Worker Node

Manual Scheduling

Create Pod

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nodeName = worker-node-1

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Worker Node 1

No decision is made by the Scheduler because you already chose the node.

Comparison Table

Automatic Scheduling	Manual Scheduling
Scheduler selects the node	User selects the node
Default behavior	Requires nodeName
Best for most workloads	Used for special cases
Uses Scheduler	Bypasses Scheduler

Advantages of Automatic Scheduling

- Easy to use.
 - Automatically balances workloads.
 - Selects the best available node.
 - Recommended for most applications.
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Advantages of Manual Scheduling

- Full control over Pod placement.

- Useful for testing.
 - Can target nodes with special hardware.
 - Helpful for specific workloads.
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Disadvantages of Manual Scheduling

- You must know the node names.
 - If the selected node is unavailable, the Pod may remain in the **Pending** state.
 - Not commonly used in production unless there is a specific need.
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Common Commands

View Nodes

kubectl get nodes

Example Output

NAME	STATUS
worker-node-1	Ready
worker-node-2	Ready
worker-node-3	Ready

View Pod Details

kubectl describe pod nginx-pod

This shows which node the Pod is running on.

View Pods with Node Information

kubectl get pods -o wide

Example Output

NAME	READY	STATUS	NODE
------	-------	--------	------

nginx-pod 1/1 Running worker-node-1

Easy Revision

Scheduling

- Decides where a Pod should run.
- Done by the Kubernetes Scheduler.
- Scheduler selects the best Worker Node.
- Automatic by default.

Manual Scheduling

- User selects the Worker Node.
 - Uses the nodeName field in the Pod YAML.
 - Scheduler is bypassed.
 - Used only when specific node placement is required.
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Easy Memory Trick ★

Scheduling

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Scheduler decides

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Best Worker Node

Manual Scheduling

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User decides

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Specific Worker Node

Real-Life Memory Trick

- 🤖 Automatic Scheduling = GPS chooses the best route for you.
 - 🕒 Manual Scheduling = You choose the exact route yourself.
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KCNA Interview Answer

What is Scheduling?

Scheduling is the process where the Kubernetes Scheduler selects the most suitable Worker Node to run a Pod based on available resources and scheduling rules.

What is Manual Scheduling?

Manual Scheduling is a method where the user specifies the Worker Node using the nodeName field in the Pod specification, bypassing the Kubernetes Scheduler